

Multiple Choice (10 marks)

1. Which types of functions grow the slowest?
 - (a) $O(\log e^N)$
 - (b) $O(\log N)$
 - (c) $O(\log \log N)$
 - (d) $O(N)$
2. In Python 3, what is `['a', 'Chemistry', 0, 1][-3]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
3. What is the value of this python expression: `4 + 3 % 2`?
 - (a) 1
 - (b) 6
 - (c) 4
 - (d) 5
4. In Python 3, which of the following function sets the integer starting value used in generating random numbers?
 - (a) `choice(seq)`
 - (b) `randrange ([start,] stop [,step])`
 - (c) `random()`
 - (d) `seed([x])`
5. In python 3, which of the following is floor division?
 - (a) `/`
 - (b) `//`
 - (c) `%`
 - (d) `|`

True / False (10 marks)

Answer True or False

6. True or False: The function `sorted('ab')` returns a new list containing the characters of the string 'ab' sorted in ascending order.
7. True or False: In Python 3, the function `max([1, 2, 3, 4])` returns the value 1, which is the smallest number in the list.
8. True or False: In Python 3, the expression `['a', 'Chemistry', 0, 1][1]` evaluates to 0.
9. True or False: In Python 3, the `strip()` function removes all leading and trailing whitespace from a string.
10. True or False: $O(n^2)$ represents a larger asymptotic growth rate compared to $O(n)$, $O(1)$, and $O(\log n)$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of converting the decimal number 231 into its hexadecimal representation. Explain each step and the significance of the hexadecimal system in computer science.
12. Discuss the characteristics of phishing attacks and evaluate why an email from your bank asking you to call a known number may not be a phishing attempt.

End of Paper